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Data Science and Information Technologies

MSc THESIS

MicroRNA target prediction with novel machine learning methods beyond quadratic attention

Rafail N. Adam

Supervisors: Martin Reczko, Professor BSRC-Alexander Flemming

Code: https://github.com/RafailTn/msc-thesis/tree/main

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ΑΘΗΝΑ

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Rafail N. Adam
S.N.: 7115152400009

SUPERVISORS: Martin Reczko, Professor BSRC-Alexander Flemming

EXAMINATION COMMITTEE:

Martin Reczko, Professor BSRC-Alexander Flemming
Artemis Hatzigeorgiou, Professor University of Thessaly
Alexandros Dimopoulos, Assistant Professor Harokopio University

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ΕΠΙΒΛΕΠΟΝΤΕΣ: **Martin Reczko**, Καθηγητής BSRC-Alexander Flemming

ΕΞΕΤΑΣΤΙΚΗ ΕΠΙΤΡΟΠΗ:

Martin Reczko, Καθηγητής BSRC-Alexander Flemming
Artemis Hatzigeorgiou, Καθηγήτρια Πανεπιστήμιο Θεσσαλίας
Αλέξανδρος Δημόπουλος, Επίκουρος Καθηγητής Χαροκόπειο Πανεπιστήμιο

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ABSTRACT

MicroRNAs are small, non-coding RNA molecules that partake in a variety of regulatory mechanisms, with the most well-studied being their role in the suppresion of translation of mRNAs via binding to sequences called MicroRNA Response Elements (MREs). They are thus implicated in a wide range of mechanisms in physiology and their disregulation can be decisive for disease states. In the current age of multi-omics analyses and personalized medicine, miRNAs are discovered via high-throughput sequencing experiments such as small RNA-seq and their regulatory roles can be studied via experiments such as AGO-eCLIP and AGO-CLASH that rely on the quantification of miRNA-mRNA hybrids. Given that a miRNA can target several different mRNA molecules, it is often complex to discern the regulatory effect of many microRNAs from a single high-throughput sequencing experiment which necessitates the use of advanced computational analyses for this purpose. To that end, several machine learning and deep learning models have emerged with the goal of miRNA target prediction. Such models aim either to predict valid miRNA-MRE pairs, or make predictions at the gene level, either finding possible MREs within the entire gene/mRNA sequence or using this information to predict changes in gene expression with the integration data from additional high-throughput sequencing experiments regarding the transriptome. The goal of this thesis is to develop a machine/deep-learning model that, given pairs of microRNAs and MREs predicts which ones would interact. For this purpose, AGO2-eCLIP data were utilized for training and evaluation of several models leading to the development of an weighted ensemble of machine learning models that is able to discern valid from invalid such pairs with an Average Precision score comparable to other state of the art models trained on the same task. The output of the model also allows for advanced interpretability via providing scores for the contribution of each feature to the final prediction. The code and instructions to reproduce the results of this thesis are publicly available at https://github.com/RafailTn/msc-thesis/tree/main.

SUBJECT AREA: Bioinformatics / Computational Biology / Machine Learning

KEYWORDS: miRNA Target Prediction, Machine Learning, AGO2-eCLIP, MRE, Model Interpretability

ΠΕΡΙΛΗΨΗ

Τα microRNA είναι μικρά, μη κωδικοποιητικά μόρια RNA που συμμετέχουν σε ποικίλους ρυθμιστικούς μηχανισμούς, με πιο καλά μελετημένο τον ρόλο τους στην καταστολή της μετάφρασης των mRNA μέσω της σύνδεσής τους σε αλληλουχίες που ονομάζονται Στοιχεία Απόκρισης microRNA (MREs). Ως εκ τούτου, εμπλέκονται σε ένα ευρύ φάσμα φυσιολογικών μηχανισμών και η απορρύθμισή τους μπορεί να είναι καθοριστική για παθολογικές καταστάσεις. Στη σημερινή εποχή των πολυ-ομικών αναλύσεων και της εξατομικευμένης ιατρικής, τα miRNA ανακαλύπτονται μέσω πειραμάτων αλληλούχισης υψηλής απόδοσης, όπως το small RNA-seq, και οι ρυθμιστικοί τους ρόλοι μπορούν να μελετηθούν μέσω πειραμάτων όπως τα AGO-eCLIP και AGO-CLASH, τα οποία βασίζονται στην ποσοτικοποίηση των υβριδίων miRNA-mRNA

Δεδομένου ότι ένα miRNA μπορεί να στοχεύσει πολλά διαφορετικά μόρια mRNA, είναι συχνά περίπλοκο να διακριθεί η ρυθμιστική επίδραση πολλών microRNA από ένα μόνο πείραμα αλληλούχισης υψηλής απόδοσης, γεγονός που καθιστά αναγκαία τη χρήση προηγμένων υπολογιστικών αναλύσεων για τον σκοπό αυτό. Προς αυτή την κατεύθυνση, έχουν εμφανιστεί διάφορα μοντέλα μηχανικής μάθησης και βαθιάς μάθησης με στόχο την πρόβλεψη στόχων των miRNA. Τέτοια μοντέλα στοχεύουν είτε στην πρόβλεψη έγκυρων ζευγών miRNA-MRE, είτε στην πραγματοποίηση προβλέψεων σε επίπεδο γονιδίου, είτε ανευρίσκοντας πιθανά MREs εντός ολόκληρης της αλληλουχίας του γονιδίου/mRNA, είτε χρησιμοποιώντας αυτές τις πληροφορίες για την πρόβλεψη αλλαγών στη γονιδιακή έκφραση με την ενσωμάτωση δεδομένων από πρόσθετα πειράματα αλληλούχισης υψηλής απόδοσης που αφορούν το μεταγράφωμα.

Ο στόχος αυτής της διατριβής είναι η ανάπτυξη ενός μοντέλου μηχανικής/βαθιάς μάθησης το οποίο, δεδομένων ζευγών microRNA και MREs, θα προβλέπει ποια από αυτά αλληλεπιδρούν. Για τον σκοπό αυτό, χρησιμοποιήθηκαν δεδομένα AGO2-eCLIP για την εκπαίδευση και την αξιολόγηση αρκετών μοντέλων, οδηγώντας στην ανάπτυξη ενός σταθμισμένου συνόλου (weighted ensemble) μοντέλων μηχανικής μάθησης, το οποίο είναι σε θέση να διακρίνει τα έγκυρα από τα μη έγκυρα ζεύγη με σκορ Μέσης Ακρίβειας (Average Precision) συγκρίσιμο με άλλα μοντέλα αιχμής που έχουν εκπαιδευτεί στην ίδια εργασία. Το αποτέλεσμα του μοντέλου επιτρέπει επίσης προηγμένη ερμηνευσιμότητα, παρέχοντας βαθμολογίες για τη συνεισφορά κάθε χαρακτηριστικού (feature) στην τελική πρόβλεψη.

ΘΕΜΑΤΙΚΗ ΠΕΡΙΟΧΗ: Βιοπληροφορική / Υπολογιστική Βιολογία / Μηχανική Μάθηση

ΛΕΞΕΙΣ ΚΛΕΙΔΙΑ: Πρόβλεψη στόχων miRNA, Μηχανική Μάθηση, AGO2-eCLIP, MRE, Ερμηνευσιμότητα Μοντέλων

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1. INTRODUCTION

1.1 MicroRNAs

MicroRNAs are the most well characterized class of non-coding RNA molecules, most commonly between 18-22 nucleotides long, that act mainly to regulate the post-transcriptional landscape of the cell, functioning as small guide RNAs for proteins of the Argonaute (AGO) family. Typically, miRNAs bind to the target mRNA sequence via their seed region which covers positions 2-8 in their sequence. Perfect complementarity leads to AGO mediated cleavage of the target with the Argonaute proteins functioning as endonucleases. Consequently, the translation of the target mRNA molecule halts and it is further targeted by other cellular RNA nucleases that lead to its complete degradation. In that manner, they are able to function in a vast variety of regulatory networks that control cellular behaviour. MicroRNAs are named based on the miRBase scheme whereby a 3-letter code denotes the species the miRNA was discovered in (e.g hsa: Homo sapiens), followed by miR in the case of mature sequences (or mir for precursor ones) and finally a number that conveys when the molecule was discovered/published with respect to the other entries (AMBROS *et al.* 2003; Kozomara & Griffiths-Jones 2011). As a rule, miRNAs from different organisms that are orthologous share the same name after the species identifier, whereas paralogous sequences have a single-letter suffix that distinguishes them and hairpin loci that give rise to the same mature miRNA are specified by a single-digit suffix. Finally, in cases where two different molecules are excised from the same precursor, a 5p or 3p suffix is used. It is also worth noting that with the advent of advanced high-throughput sequencing techniques, it has become plausible to detect isoforms of miRNAs. Such molecules differ from the canonical miRNA sequence either in terms of length of their 5’ and/or 3’ ends or because their sequence has been modified via processes like A-to-I or C-to-U editing. These modifications subsequently change how they are loaded to AGO and how they interact with target MREs (Tomasello *et al.* 2021).

1.1.1 Biogenesis

The transcription of a miRNA gene is performed by RNA Polymerase II or RNA Polymerase III and leads to the production of a precursor called pri-miRNA which is longer in length than the mature molecule and has the structure of a stem-loop with single-stranded overhangs extending from its 3’ and 5’ ends. It is necessary that a stem of a pri-miRNA molecule is of specific size, its loop is at least ten base pairs long and its ends can produce single-stranded RNAs, so that it can be processed into a mature miRNA (Zeng & Cullen 2003; Zhang & Zeng 2010). The maturation process starts with a protein complex comprising of an endoribonuclease called Drosha and a protein called DGRC8. DGRC8 is a protein that binds double-stranded RNA, in this case the stem of the pri-miRNA, stabilizing its interaction with Drosha. The cleavage of the pri-miRNA by Drosha is also guided by DGRC8 which recognizes the region 11 base pairs downstream from its binding site.

The product is pre-miRNA, a molecule that is 60-70 base pairs long and that is exported from the nucleus by exportin 5. An alternative origin for miRNA genes is introns of protein coding genes, such genes are also called mirtrons. Due to the fact that the pri-miRNA molecules originate from introns,they do not have the appropriate structure needed in order to be processed by Drosha and DGRC8. A protein called DRB1 solves the lariat structure of intronic RNA and pre-miRNA is the immediate product of this process. Another example of Non-canonical miRNA biogenesis includes 7-methyl Guanosine (m7G)-capped pre-miRNA, which are exported to the cytoplasm via Exportin 1 without being cleaved by Drosha first. All pre-miRNAs are recognized by a type 3 RNAse called DICER and TRBP which is a protein that binds the stem of the pre-miRNA. DICER can then cleave the pre-miRNA molecule into two leading to products of 22 base pairs long, one of which will be loaded into the RISC complex that is comprised of DICER, TRBP, AGO, PACT, and TNRC6A-C (O'Brien *et al.* 2018).

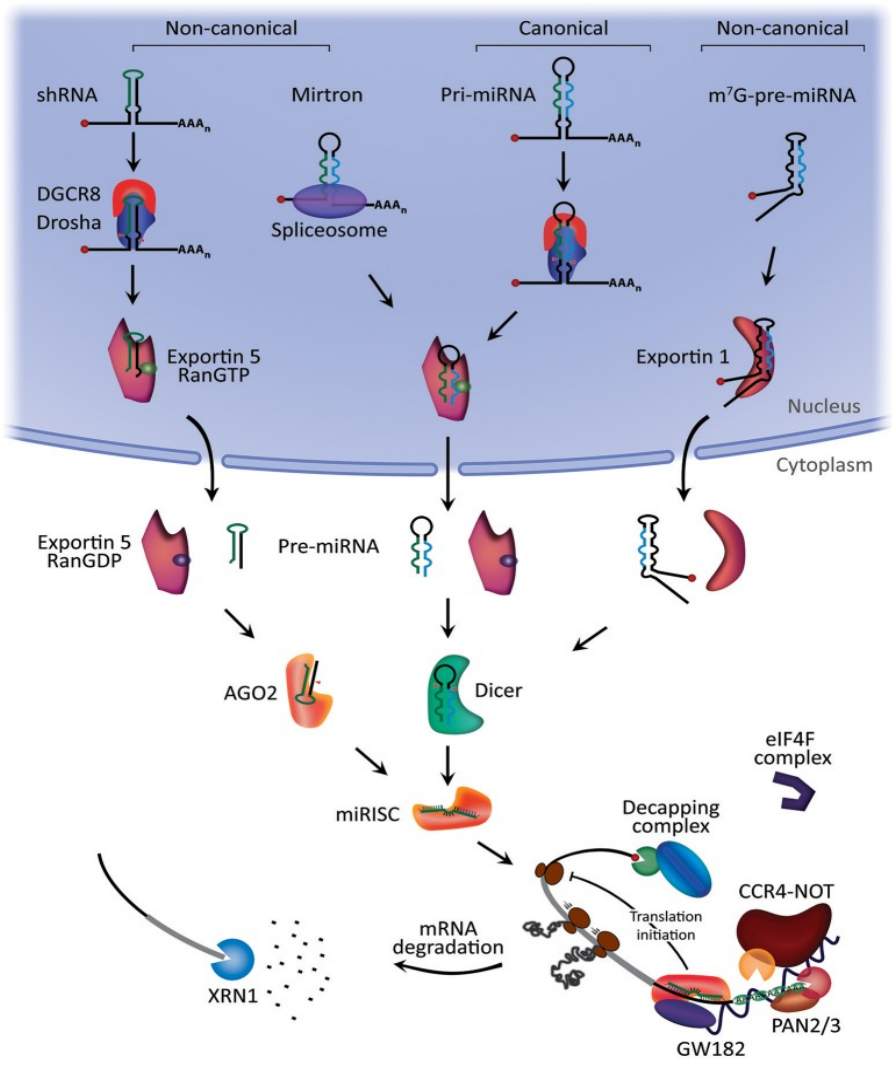


Figure 1.1: The steps of miRNA biogenesis. (O'Brien *et al.* 2018)

1.1.2 Target Binding

The strand that is loaded is to be used as a guide RNA for RISC, via hybridizing with the mRNA. The dsRNA is mainly formed between the seed region of the miRNA (nucleotides 1-8) and the 3'UTR of the mRNA, with additional, downstream base pairing aiding the interaction. Given perfect complementarity, the endonuclease activity of AGO2 is induced and the mRNA is cleaved. At the same time, proteins of the GW182 complex are recruited in order to act as scaffolds for necessary effector proteins such as PAN2-PAN3 and CCR4-NOT complexes that function as Poly-A Deadenylases that remove the protective 3' polyA tail, given the interaction between tryptophan from W repeats of the GW182 complex and the PABC complex on the mRNA has been established. Finally, removal of the 5'cap of the mRNA from DCP2 takes place followed by 5'-3' degradation from the exonuclease XRN1.

Most of the miRNA:MRE complexes in the cell are not perfectly complementary and additional 3' base pairing plays an important role for their stabilization. The regulatory efficiency scales with the number of seed matches according to the following order (Grimson *et al.* 2007):

1. 8mer: 8 matches within the seed region (1-8)
2. 7mer m8: 7 matches within the seed region (2-8)
3. 7mer A1: 7 matches within the seed region with the first nucleotide being an Adenine (1-7)
4. 6mer: 6 matches within the seed region (2-7)
5. 6mer offset: 6 matches within the seed region (3-8)
6. 5mer: 5 matches within the seed region (2-6)

As mentioned above, additional base pairing can enhance the stability of the complex and is classified as supplementary binding if it does so without being necessary for the regulatory role of the miRNA, or as compensatory if it compensates for mismatches within the seed region. Finally, there is a seedless mode of pairing for miRNAs that has been described in C.elegans which likely occurs due to AU rich sequences in the seed region. Regulatory efficiency is also affected by relationships between different binding sites and binding events within the same mRNA molecule. Adjacent sites for different miRNAs and repetitive sites for the same miRNA can both function in a cooperative manner. Functional cooperativity refers to the event whereby many different binding events have a cumulative effect, whereas binding cooperativity to the physical interaction of different RISC complexes supporting each other in the binding of their target (Diener *et al.* 2024).

1.1.3 Additional mechanisms of action

Binding of miRNAs on MREs can also increase the efficiency of translation instead of suppress it. A protein called FXR1 is necessary for this phenomenon, which in combination with RISC may bind specific sequence motifs (such as AU Rich Elements, AREs) or other sequences in the 5'UTR leading to activation of translation. MicroRNAs can also regulate gene expression via utilizing other, albeit less studied, mechanisms. For example, RISC complexes can be transported in the nucleus in order to exert regulation on transcription. In order for this to occur, they have to be imported from the cytoplasm via importin 8, importin α/β or XPO1. Within the nucleus, these complexes can bind to DNA regulatory elements affecting the binding of transcription factors and of other regulatory proteins (such as chromatin remodelling complexes) on them, thus having an impact on the transcription of related genes (Kim *et al.* 2008; Chaluvally-Raghavan *et al.* 2016; Xiao *et al.* 2017; Guo *et al.* 2021). A noteworthy example is that of miR-30b-5p which upon starvation, translocates to the nucleus in order to bind regulatory elements known as CLEAR elements that are known to bind the TFEB transcription factor. In that manner, it suppresses the transcription of these genes and inhibits lysosomal biogenesis and the autophagic flux (Guo *et al.* 2021). Additionally, RISC has been shown to be able to bind to mRNAs within the nucleus and induce their degradation while AGO and Drosha also participate in the splicing of premature mRNAs (O'Brien *et al.* 2018).

1.1.4 Roles in Physiology and Disease

Given their diversity regarding both sequence and mechanisms of action, miRNAs partake in many physiological and pathological states. Due to their capability of regulating multiple genes in a variety of different manners, they are important components in developmental processes that require the fine-tuning of gene expression at precise time intervals, with a prominent example being that of miR-196. Anterioposterior polarity is mediated by the SHH protein in forelimb and hindlimb buds and its expression requires Retinoic Acid (RA) signalling induced Hoxb8 expression. Hoxb8 is a transcription factor protein that is upregulated as an immediate-early response to ectopic RA administration to forelimb but not to hindlimb buds due to a specific miRNA, miR-196, that inhibits the translation of Hoxb8 in a tissue specific manner (Hornstein *et al.* 2005). MicroRNAs are also able to regulate protein synthesis spatiotemporally within the cell, localizing to specific subcellular compartments before exerting their effects (Corbin *et al.* 2009).

Deregulation of miRNA expression can lead to or deteriorate a big number of disease states ranging from congenital defects to autoimmune diseases, neurological disorders and many types of cancer. For instance, Adam *et al.* 2009 showed that miR-200 family expression in UMUC3 cells lead to increase in E-Cadherin expression and elevated sensitivity to EGFR inhibitors. MiRNAs can also be excreted from one cell to be taken in by a target via exosomes. Lin *et al.* 2019 demonstrated that miR-21 is excreted from bladder cancer cells via exosomes that are taken in by macrophages in order to activate the PI3K/AKT pathway and promote cancer progression by polarizing them towards the M2 phenotype. In addition, miRNAs can be detected in fluid samples from patients and can thus be used as biomarkers for specific diseases. For instance, Sapre *et al.* 2016 showed that a combination of 6 miRNAs from urine are able to predict Urothelial Carcinoma of the Bladder (UCB) patients with a high AUC score (0.85). Finally, miRNAs can be used as targets for therapeutics that either induce or reduce their expression. An example is miR-222 which has elevated expression levels in cardiomyocytes after specific types of exercise as well as following exercise rehabilitation after heart failure. This elevated expression leads to targeting of p27, HIPK-1, HIPK-2, and HMBOX1 which promotes cardiomyocyte hypertrophy, proliferation, and survival (Ding *et al.* 2017).

1.1.5 High-throughput sequencing methods for miRNA research

In the era of multiomics studies, it has become prevalent to study the totality of the miRNA part of the transcriptome as well as their targets via the use of high-throughput sequencing techniques. Such techniques allow for the detection and quantification of all the miRNA molecules in a tissue as well as of the complexes that these form with MREs. The most widely used one is small RNA-seq which includes isolation of RNA, cDNA library construction and sequencing. During RNA isolation, the total RNA might be extracted from a tissue, optionally followed by enrichment methods such as length separation or binding of small RNAs to to specific proteins. Library construction involves a reverse transcription step followed by ligation of adapters and/or polyadenylation (Benesova *et al.* 2021). The following image depicts the double ligation small RNA-seq protocol:

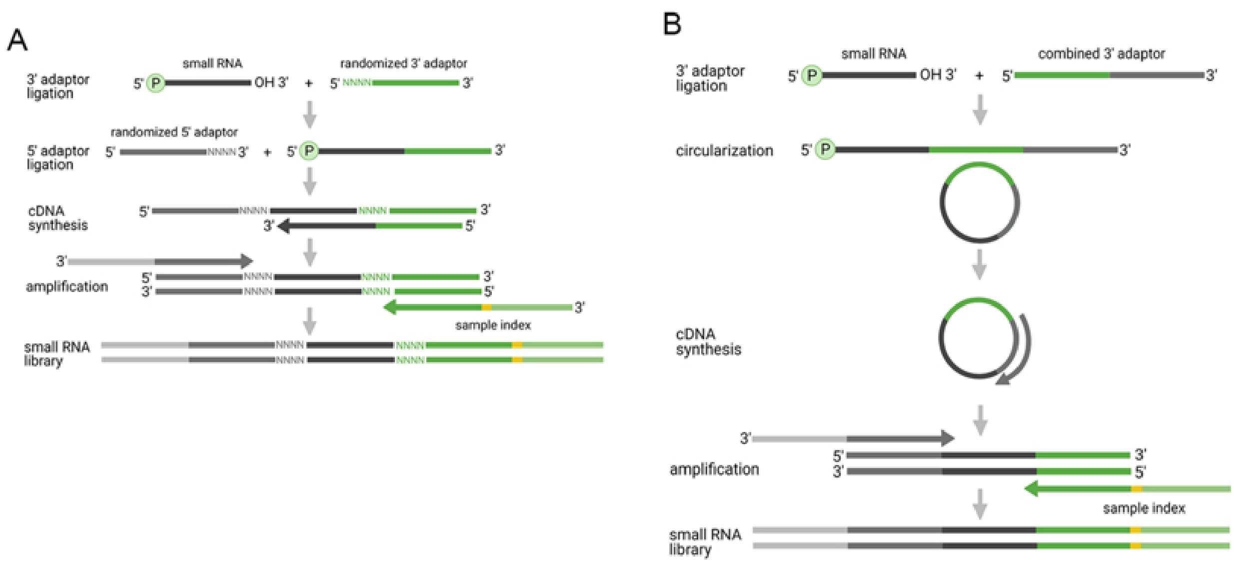


Figure 1.2: Double ligation small RNA-seq. (Benesova *et al.* 2021)

Types of experiments that capture and quantify miRNA:MRE complexes are used to study the binding sites of miRNAs and include AGO-eCLIP and AGO-CLASH. The latter involves the following steps (A *et al.* 2013):

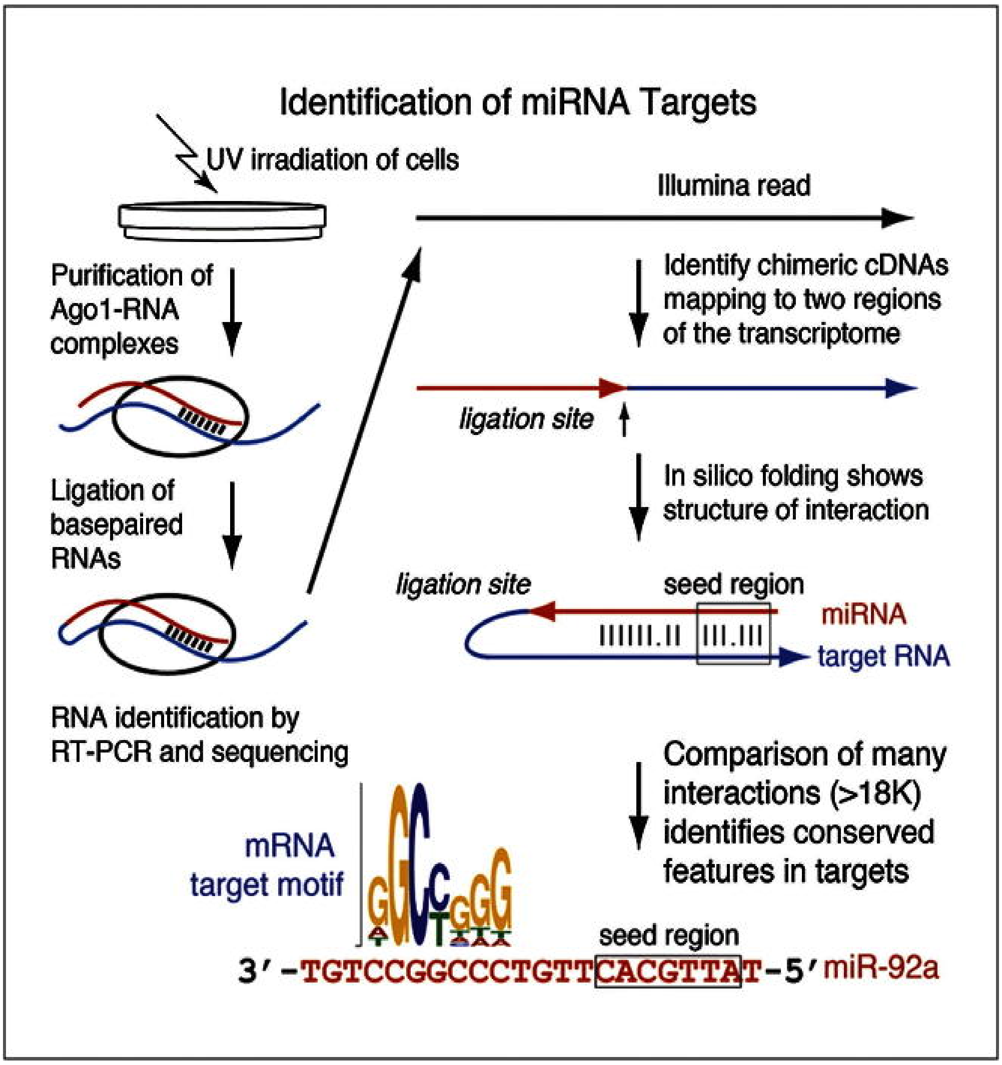


Figure 1.3: AGO-CLASH. (A *et al.* 2013)

Before cross-linking via UV irradiation, AGO expression is induced via the use of Doxicyclin, where the concentration is monitored in order for AGO to be close to endogenous levels. Despite that, differences between endogenous expression do occur and with them, the risk of bias regarding specific interactions. After cross-linking, hydrolysis of interacting RNA molecules takes place followed by ligation of the hybridized miRNA:MRE duplex which creates hybrid molecules to be reverse transcribed, amplified and sequenced. AGO-eCLIP follows a different procedure that does not require modifications to the endogenous expression. During AGO-eCLIP, the RISC:MRE complex is cross-linked using UV irradiation and the RNA is fragmented in the same manner as with AGO-CLASH, then immunoprecipitation, using antibodies that bind the AGO protein, follows and ultimately ligation of the mRNA 5'-P to the miRNA's 3'-OH group (Manakov *et al.* 2022). Due to the immunoprecipitation step, no exogenous induction of expression is needed, eliminating the aforementioned bias and providing an image that is more physiologically relevant.

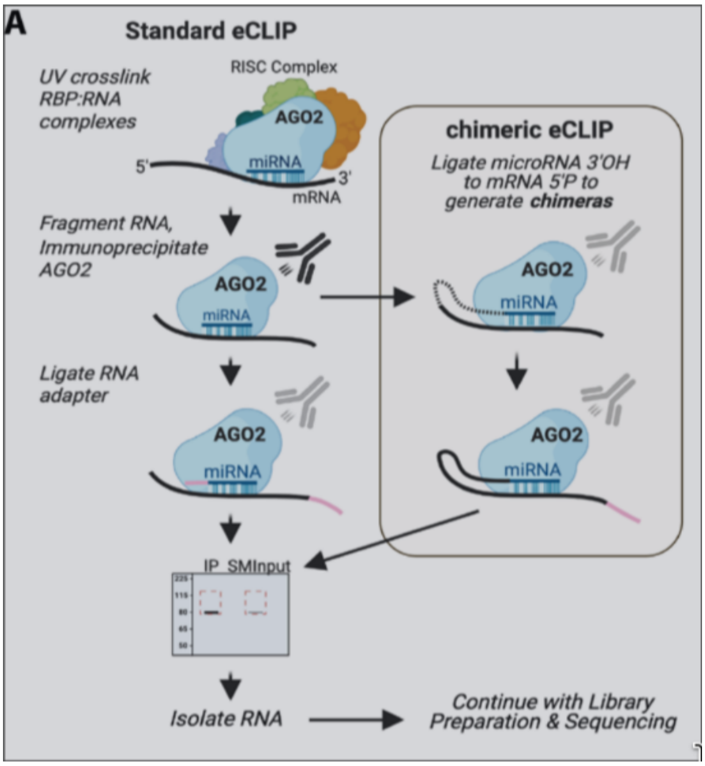


Figure 1.4: AGO-eCLIP. (Manakov *et al.* 2022)

There is also a family of techniques used to quantify the regulatory effect of miRNAs on their targets, which includes Degradome Sequencing, Parallel Analysis of RNA Ends and Genome-Wide Mapping of Uncapped and Cleaved Transcripts (Gregory *et al.* 2008; Addo-Quaye *et al.* 2008; German *et al.* 2009). In these experiments, the polyA transcripts are isolated from the tissue of interest, then ligation of the first adapter occurs only on transcripts with 5' monophosphate (which are the ones that have lost their 5'm7G cap due to miRNA mediated decay), followed by reverse transcription, 3' adapter ligation, PCR and sequencing. The result is depicted as a peak on an mRNA transcript where the x-axis shows the sequence the miRNA targeted and the y-axis the frequency of 5' ends found at this position.

High-throughput sequencing techniques produce enormous amounts of data regardless of their end goal. Commonly, the immediate result from the sequencing instrument involves files that are called FASTQ files which include entries of all the sequenced molecules along with a quality score that conveys the reliability of the base that was sequenced at a given position. Each entry of a FASTQ file follows the format below:

1. 1st Line: @ character followed by an identifier for the entry
2. 2nd Line: Raw nucleotide seq
3. 3rd Line: + character optionally followed by the sequence identifier again
4. 4th Line: quality values for the sequence

It is worth noting that there is a number of different sequencing instruments that can be used for the same protocol but utilize different biochemistry principles in order to perform sequencing at such large scales. Due to these differences, the end result will be different even if one were to sequence the exact same samples and as such specific methods have been developed to mitigate this issue and ensure reproducibility across studies. Such methods are necessary when one wants to combine results from a number of different studies that have performed these experiments and the process called Batch Effect Correction. Finally, even within the same experiment it is heavily suggested that there are more than one replicates for the same sample in order to account for technical variation in the measurements.

Depending on the sequencing protocol and the end goal, these files are processed following different pipelines which include the usage of a series of computational tools. A common step regardless of the protocol is the Quality Control step (QC) in which various plots are produced in order to examine the integrity of the sequenced molecules based on the quality scores themselves as well as knowledge regarding the expected nucleotide and length distributions. Depending on the results, the next step involves filtering based on specific thresholds and optional removal of the adapter sequences followed by another QC step to ensure integrity has been maintained.

The next step is to map the sequences in the filtered file on the human genome/transcriptome and it is the first source of heterogeneity pertaining to such pipelines. Tools that perform this mapping are called Aligners and there is a big number of such tools each of which excels in specific circumstances. For small RNA-seq, two aligners that are commonly used are Bowtie2 and BWA-MEM (Langmead *et al.* 2009; H. Li 2013), that achieve high speed alignment of short reads on genomic sequences. The choice of the alignment tool as well as that of its alignment parameters are vital for the quality of the end result which necessitates their inclusion in the pipeline for reproducibility. A vital aspect of this step is the genome version that is used where the reads are to be aligned, as different versions lead to differences in results. Most studies use hg38 which is considered the most up to date and complete version, although with deeper and more accurate sequencing more recent versions like the Telomere to Telomere version have emerged which might come to replace hg38, especially for the analysis of repetitive regions such as centromeres, telomeres and ribosomal array sequences. For AGO-eCLIP and AGO-CLASH, a typical aligner would not suffice as the reads that are sequenced are chimeric molecules, consisting partly of miRNA and of mRNA fragments. Thus, specific tools have been developed that can appropriately align each part, with an example being Hyb (Travis *et al.* 2014). Hyb begins by aligning the reads either to the genome or to a known database of molecules (ideally mature mRNAs) in order to identify reads that are non-contiguously aligned and then utilizes folding algorithms to annotate the recovered duplexes.The end result is a list of RNA:RNA hybrids.

1.2 miRNA Target Prediction Algorithms

Several methods have been developed over the years that utilize the data produced by the aforementioned experiments in order to achieve several goals related to elucidating miRNA biology. In general, predictive algorithms are separated in terms of the type of algorithms they use in order to make the prediction as well as the prediction task itself.

- Algorithms
  - Heuristic
  - Physics Based
  - Machine/Deep Learning
- Task
  - Site Level
  - Gene/Functional Level
  - Pathway Level

1.2.1 Site Level predictors

In terms of algorithms, a tool is characterized as Heuristic if it makes predictions using pre-set rules that are based on already discovered interactions (Lewis, Shih, *et al.* 2003; Lewis, Burge, *et al.* 2005). The first release of TargetScan was heuristics based, as it searched for perfect seed complementarity in the 3'UTR of each target provided, extended it to include as many matches as possible and then used software like RNAFold to check for additional complementarity in the miRNA 3'UTR with the next 35 nucleotide of the mRNA (Hofacker *et al.* 1994) and assigned a Z score that is the basis for the final predictions. Such tools are considered the simplest form of prediction tools but function as a good baseline for comparison with more complex models.

Physics based tools may disregard the biology behind the task and focus on expressing the problem as an energy minimization one when it comes to MRE prediction although more recent ones aim to strike a balance between using some heuristics for narrowing down some predictions in combination with energy minimization. A fine example of this is IntaRNA (Busch *et al.* 2008; Mann *et al.* 2017; Gelhausen *et al.* 2019; Wright *et al.* 2014; Miladi *et al.* 2019; Raden *et al.* 2020) which is a rather advanced tool that provides a lot of different modes and was used as part of the feature extraction in this thesis.

The most relevant tool regarding target site prediction for the scope of this thesis is mirBind (Klimentová *et al.* 2022) as it has been tested on the dataset from Manakov *et al.* 2022.The *miRBind* model employs a deep learning architecture, built primarily upon Convolutional Neural Networks (CNNs) and Residual Networks (ResNets), to predict microRNA (miRNA) to target site binding without relying on traditional, rigid seed-matching heuristics. Instead of processing sequence strings independently in one dimension, the network encodes the input pair, typically a 20-nt miRNA and a 50-nt target sequence, into a two-dimensional pairwise representation array. This 2D matrix explicitly captures all possible nucleotide matches and mismatches across the two sequences. The core feature extractor consists of a stack of multiple convolutional or residual blocks. Each block sequentially applies 2D convolution operations, Leaky ReLU activations, batch normalization, max pooling, and dropout regularization to learn complex, non-linear interaction rules directly from the spatial alignment data. Following the convolutional stages, the resulting high-level feature maps are flattened and fed into a series of dense (fully connected) layers. The network terminates at a single output neuron with a sigmoid activation function, which yields a continuous probability score indicating the likelihood of a functional binding event between the miRNA and the target transcript.

1.2.2 Functional and Pathway Level predictors

The advancement of machine and deep learning algorithms has paved the way for the development of models that circumvent the need for feature engineering and go beyond site level prediction. Heuristic and Physical approaches as those described above, as well as classical machine learning algorithms, need to take as input a number of features among which they try to calculate weights and correlations so as to create a function that separates the samples into predifined categories or fit a regression line on a multi-dimensional distribution. This requires prior processing of the samples so as to extract such meaningful features utilizing domain knowledge for the specific task. Modern deep learning models do not require such intensive preprocessing as they are developed in order to process the nucleotide sequence directly and find more abstract patterns that correlate with the desired output. The first and simplest type of architecture that has been used is Convolutional Neural Networks (CNNs). CNNs allow for the processing of the nucleotide sequence outside of the tabular limitations of classical Machine Learning by reading it in predifined windows along which mathematical operations are performed with the goal of creating a feature map whereby pooling is performed in order to reduce overfitting and make the model translation invariant. The final part usually consists of a classification layer that is used to output a probability for a specific class.

Some modern approaches like the one from Zacharopoulou *et al.* 2024 employ such architectures combining them with ones like Gated Recurrent Units (GRUs) in order to extract a unified score across an entire gene and use it as a feature for a meta learner that is trained on expression data to predict functional binding which goes beyond site level predictions. The most recent and advanced models perform what is called Pathway Level prediction, by taking advantadge of site/gene level predictors and Graph Neural Networks (GNNs), they predict pathway level interactions between mRNAs-miRNAs-Competing Endogenous RNAs (ceRNAs), making them more relevant for identifying which of these pathways are crucial for disease states (Chiu *et al.* 2015; Jin *et al.* 2024). Finally, modern models may also rely on Transformer based architectures which are able to treat DNA/RNA as a language and learn more intricate pattern that are not necessarily in proximity in the input sequence. Examples of these include the vanilla Transformer, Titans and the Mamba architecture which is not transformer based but has shown promise in DNA related tasks (Behrouz *et al.* 2024; Gu & Dao 2024)

1.2.3 Main Goal

The main goal of this thesis is to explore the use of architectures and algorithms that aim to surpass Transformers in terms of either performance or explainability. Thus,the Mamba architecture was employed and its results were compared to those of a vanilla Transformer deep neural network. Additionally, simpler machine learning models and their ensembles were employed in order to further examine the tradeoff between their higher degree of explainability and (potentially) speed and their performance on the Manakov *et al.* 2022 dataset.

2. METHODS

2.1 Data preprocessing

The dataset used for training and testing was taken from Sammut *et al.* 2025 and contains preprocessed files from experimental results produced by Manakov *et al.* 2022; Hejret *et al.* 2023. These include, apart from the DNA sequences of the miRNA and the MREs, information about the miRNA family, the isomiR that the molecule is identified as, the hg38 genomic coordinates of the MRE, two types of conservation vector scores (genePhastCons100way and genePhyloP) in lists, what region of the mRNA is targeted (exon, intron, 3'UTR, 5'UTR and boundaries between them), gene cluster ID (which is used for negative sample generation) and labels (1 for positives, 0 for negatives). The dataset is balanced in terms of positives and negatives not only in its totality but also within miRNA families. This is crucial, as different families may bind in different manners and with different criteria and thus it is important that any potential model learns all such patterns.

The steps for negative sample generation according to Sammut *et al.* 2025 are as follows:

1. Cluster MREs of >90% similarity
2. For a given miRNA family:
   (a) Identify clusters containing targets of that family
   (b) Choose an MRE from a cluster that does not overlap with the aforementioned ones
   (c) Assign it to each miRNA of that family to create the negative sample pair

Another bias that typically occurs apart from the skewed positive to negative ratio, has to do with the tissue of origin. More specifically, given that the dataset used comes from a single cell line (in this case HEK-293), it is essentially certain that there will be a sampling bias since tissue specific expression is not uniform. Therefore, specific miRNAs and targets are expressed more than others and are more likely to be captured by the experimental procedure. Adding to that, specific interactions of miRNA and MREs might exist solely due to this tissue specific expression pattern and conversely, specific negative samples might be detected as valid interactions if one was to use another tissue for the same experiment, due to different expression patterns. In order to mitigate this, data from TarBase (SETHUPATHY *et al.* 2006) were extracted in order to assess which of the interactions in the Manakov dataset were truly negative, based on their absence in experimental data, and which are false negatives that could hinder the models ability to learn binding patterns.

The latest version of the file TarBase provides for Homo sapiens was downloaded (v9) and cleaning was performed so as to keep only valid entries with existing genomic coordinates. If an entry, that is detected as a pair of miRNA name - Gene name, had missing coordinates for the target site it was either dropped (if the same pair exists either within the same trascript region or not) or the coordinates from the entire gene were kept from Ensembl (in the case no other entry for that pair existed). The goal of this process is to keep as many valid pairs of miRNA - Target gene from TarBase along with coordinates that can be used for overlapping with the ones from the dataset later.

The pipeline to remove falsely assigned negative samples from the dataset adheres to the steps below in order to retain the initial positive to negative ratio:

For a given miRNA:

1. Check for overlapping coordinates of its target between the cleaned TarBase file and the negative samples of the dataset
2. If overlaps are found, remove the corresponding negative sample and keep track of removed samples per family
3. Remove at random as many positive samples from each family

Ultimately, less than 1% of samples were found to be False Negatives and were removed along with the same number of positives. Another aspect of the sampling bias that had to be adressed was the fact that the model has to be prevented from treating samples from rarer families as less important and therefore ignoring potential novel binding rules in the process. Sampling weights were added to the datasets that adhere to the following formula:

$$\text{Sample Weight} = \frac{\text{Sample Total}}{\text{max}(100, \text{Family Total} \times \text{Count of current Family})} \tag{2.1}$$

Sample Total refers to the total number of samples in the dataset, Family Total to the total number of miRNA families in the dataset and Count of Current Family to the number of samples corresponding to the current sample's miRNA family. The above formula ensures that samples of more abundant miRNA families get assigned with a smaller weight proportional to their family's frequency in the dataset, while rarer samples are assigned bigger weights capped based on the product in the denominator. It is worth noting that when this ratio is multiplied by the current family count, the result is equal to the same number, with small deviations for extremely rare families. This was mainly applied to the machine learning approaches that showed most promise in terms of results.

Since the goal of the model is to be able to generalize its predictions for unseen data and more specifically unseen families, data are split based on a stratified kfold cross validation scheme that takes into account the miRNA family for each split, ensuring that there are no common miRNA families across splits and that the ratio of positives and negatives is maintained. In that manner the model of each fold is trained on specific miRNA families while the ones in the validation set are used for tuning it. Furthermore, a separate test set exists that has unseen samples of families that are common with the training set along with a holdout set with completely unseen families to both the training and test set as well as to the other two files contained in the study.

Every model is tested on both of these test sets in the end of its training in order to evaluate its performance. The main metric that is used for this purpose is the Average Precision score: APS $= \sum_n (R_n - R_{n-1}) P_n$ where R_n and P_n are the Recall and the Precision at the nth threshold. APS is used because it is threshold invariant and allows for the evaluation of the model's capability to find the positive class. This is important as validation is costly and thus False Positive samples and False Negative ones must be penalized, while True Negative samples are not equally meaningful. APS in the final model was adjusted based on the weights of the samples mentioned above in order to portray a better image of the model's capability to generalize to rare/lowly expressed miRNA families and a non-adjusted APS is also reported for reference and comparison with other models.

2.2 Predictive Models

The types of algorithms that were used fall into either the classic Machine Learning category, where the library Autogluon was used (Erickson *et al.* 2020), or more recent architectures of Deep Learning models for Language Processing (i.e Transformers and Mamba).

2.2.1 Autogluon

Autogluon is an AutoML python library that aims to automate training Machine Learning models on various types of tasks that are based on tabular input or even language and images. Its logic is based on initially training many models of the same instance separately (e.g many Random Forest models) on different folds of the dataset following either the default or a user defined kfold cross validation scheme and creates an ensemble of these models as a representative of that model instance. After training all model instances, it creates a weighted ensemble model which is meant to weight the predictions of each model and find an optimal weighted sum for any future prediction. Depending on whether the number of stacks parameter has been set over 0, Autogluon performs Stacked Ensembling which is depicted in the following image:

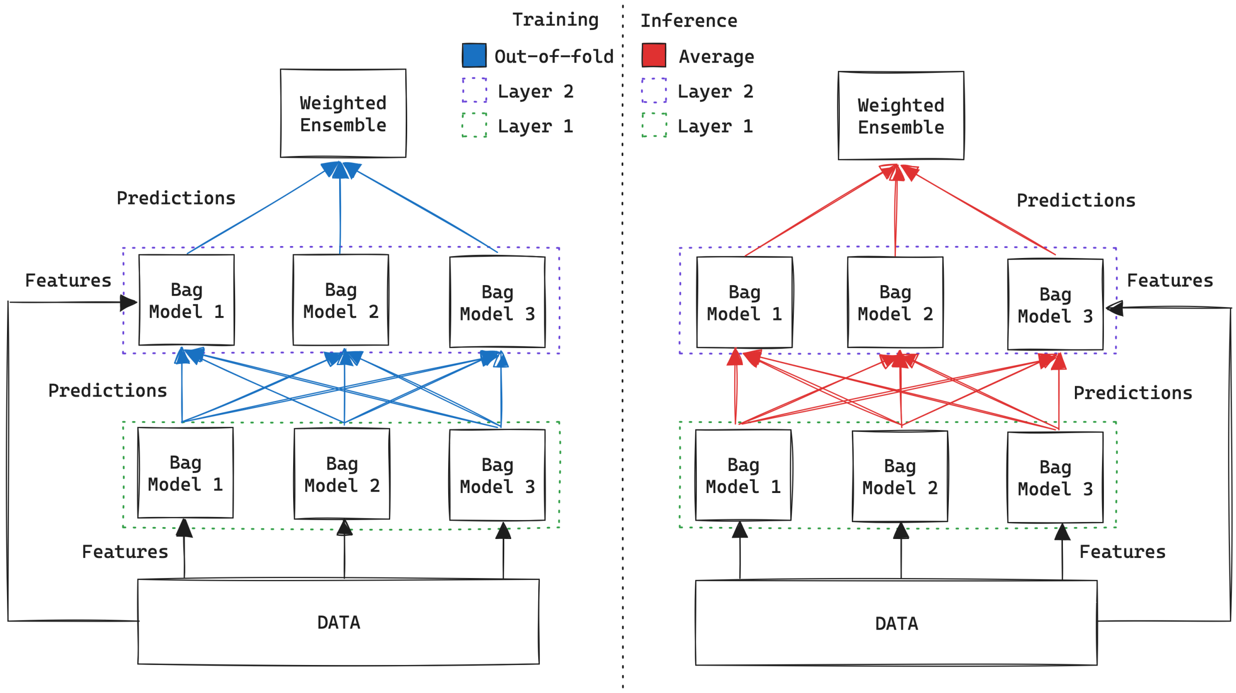


Figure 2.1: Autogluon Stacked Ensembling. (Erickson *et al.* 2020)

After the first layer of models is trained in the original featrues as described in the previous paragraph, the second layer of models (of the same instances as the first) is trained in the same manner but using as features predictions from the first layer, ultimately leading up to a second weighted ensemble. The number of layers is also a parameter that can be tuned by the user as well as the hyperparameters of each model instance. Autogluon also contains specific foundational models that are expected to have State Of The Art performance on virtually any tabular dataset. In order to avoid data leakage, Autogluon uses out of fold predictions on the stacked layers as well.

2.2.2 Transformers

Transformers were introduced in the landmark paper ”Attention is All You Need” (Vaswani *et al.* 2023). First, the input sequence is tokenized and mapped to dense vector embeddings. To inject information about the relative or absolute position of the tokens in the sequence, positional encodings, which are either learned or based on sinusoidal functions, are added to these embeddings. Through learned linear projections, the input is mapped into three distinct matrices: Queries (Q), Keys (K), and Values (V). The scaled dot-product attention mechanism, which is vital to the architecture's success, is defined by the following formula:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{2.2}$$

Rather than computing a single attention function, Transformers employ Multi-Head Attention, projecting the queries, keys, and values into multiple subspaces, performing the attention function in parallel, and concatenating the outputs. The standard Transformer encoder consists of a stack of identical layers. Each layer contains two primary sub-layers: a Multi-Head Attention mechanism and a position-wise Feed-Forward Neural Network. Every sub-layer is encompassed by a residual connection followed by layer normalization, formulated mathematically as LayerNorm$(x + \text{Sublayer}(x))$.

The Decoder follows a similar structural philosophy to the Encoder but introduces critical modifications to facilitate autoregressive sequence generation. Each identical decoder layer comprises three primary sub-layers: a Masked Multi-Head Self-Attention mechanism, an Encoder-Decoder Multi-Head Attention mechanism (often termed Cross-Attention), and a position-wise Feed-Forward Neural Network. As with the Encoder, residual connections and layer normalization encompass each of these sub-layers.

The masking in the first sub-layer is an essential modification; it mathematically prevents sequence positions from attending to subsequent positions. By setting the scaled dot-product attention scores of future tokens to $-\infty$ prior to the softmax operation, the model ensures that the prediction for position i depends exclusively on the known outputs at positions strictly prior to i.

In the subsequent Encoder-Decoder Attention sub-layer, the queries (Q) are derived from the output of the previous decoder layer, while the keys (K) and values (V) are sourced directly from the final output of the Encoder. This cross-attention mechanism allows every position in the decoder to attend over the entire input sequence, effectively grounding the generated output in the original provided context. Finally, the output of the decoder stack is passed through a linear transformation and a softmax function to produce a probability distribution over the vocabulary for the next predicted token.

2.2.3 Mamba

An alternative and highly successful paradigm has emerged in the form of Structured State Space Sequence Models (SSMs), most notably the Mamba architecture (Gu & Dao 2024). Mamba completely discards the attention mechanism, offering a fundamental architectural shift that scales linearly in time ($\mathcal{O}(N)$) with respect to sequence length, while maintaining a constant memory footprint during autoregressive generation.

At its core, Mamba is inspired by continuous-time control theory. It maps a 1-dimensional input sequence $x(t) \in \mathbb{R}$ to an output $y(t) \in \mathbb{R}$ through an implicit latent state $h(t) \in \mathbb{R}^N$.

The classical continuous-time SSM is defined by a system of linear ordinary differential equations:

$$h'(t) = Ah(t) + Bx(t) \tag{2.3}$$

$$y(t) = Ch(t) \tag{2.4}$$

To process discrete tokens (such as text), this continuous system must be discretized.

Using a timescale parameter Δ, the continuous matrices A and B are transformed into their discrete counterparts $\bar{A}$ and $\bar{B}$ (typically via the zero-order hold method). This yields a highly efficient recurrent formulation:

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t, \quad y_t = Ch_t \tag{2.5}$$

Previous iterations of SSMs (such as S4) suffered from a fatal constraint: they were Linear Time-Invariant (LTI). Because their transition matrices (B, C, and Δ) were static and fixed across all time steps, they could not selectively focus on or ignore specific tokens, making tasks like targeted information retrieval or copy-pasting nearly impossible.

Mamba resolves this by introducing a *Selection Mechanism*. It parameterizes B, C, and the step size Δ to be input-dependent functions of the current token x_t:

$$B_t = \text{Linear}_B(x_t), \quad C_t = \text{Linear}_C(x_t), \quad \Delta_t = \text{softplus}(\text{Linear}_\Delta(x_t)) \tag{2.6}$$

By allowing Δ_t to vary based on the input, the model dynamically dictates the degree to which it should integrate the current token into the hidden state (h_t) or completely ignore it. This effectively grants the recurrent model the data-filtering capabilities of an attention mechanism, without the $\mathcal{O}(N^2)$ computational penalty.

Because Mamba's parameters are time-varying, the model cannot be computed using efficient Fast Fourier Transforms (FFTs) like its LTI predecessors. To overcome this, Mamba relies on a hardware-aware parallel scan algorithm. By fusing the scan operations and materializing the expanded hidden states (h_t) exclusively in the faster, on-chip SRAM of the GPU (bypassing the slower High Bandwidth Memory (HBM) IO bottleneck) Mamba achieves training speeds competitive with highly optimized Transformers while preserving the inference efficiency of a recurrent neural network.

2.3 Feature Engineering

2.3.1 Autogluon

Although Autogluon provides models and workflows for language related data, nucleotide sequences can not be grouped in such a category without specific considerations. Firstly, nucleotide sequences do not have strictly defined words, sentences or punctuation, meaning that methods that are normally used in language processing can not be employed the same way. As such, feature engineering has to be taken care of based on domain knowledge and previous work. Due to the inspiration from MicroT-CNN and the nature of the data from miRBench, the features that were initially considered are derived from:

- MRE Conservation Scores
- Nucleotide Sequences
- Energy related scores
- miRNA:MRE Duplex Structure
- MRE secondary structure

MRE secondary structure

The tool used for MRE secondary structure prediction is Contrafold (Do *et al.* 2006), which is based on Stochastic Context-Free Grammars instead of classical physics based approaches that are typically employed for such tasks. The output of this tool is the secondary structure of the MRE in dot bracket notation which can not be used immediately as input for machine learning models and Multi-Layer perceptrons. In general, features of sequential nature which include nucleotide sequences, secondary structure and duplex structure have to be processed manually in order for them to be useful for the aforementioned models. In this case, the secondary structure was parsed and some basic features were created that include:

- Accessibility score within and outside a predicted seed region (see Duplex Structure and Energy). It is based on a crude count of paired and unpaired bases.
- Number of bulges
- Number of Hairpins
- Number of Internal Loops
- If the binding site is in a Loop
- If the binding site is in a Stem
- Length of the longest stem
- Longest stem near binding site

Duplex Structure and Energy

Two different tools were compared in terms of their outputs' ability to provide features that can be used by predictive algorithms, namely RNADuplex and IntaRNA. The output duplex structures were parsed and classified dynamically into binding type categories with the same logic as in Zacharopoulou *et al.* 2024.

RNADuplex RNAduplex makes a critical heuristic assumption: it completely ignores intramolecular base pairing. By assuming the two RNA strands are essentially unstructured prior to binding, it restricts the dynamic programming search space exclusively to interactions between the two distinct strands. This effectively reduces the thermodynamic folding problem to a variant of local sequence alignment, plummeting the time complexity to $\mathcal{O}(N \times M)$ and making it ideal for rapid, large-scale genomic screening.

The algorithm utilizes a 2D dynamic programming matrix to compute the MFE of hybridization. It relies on the Turner nearest-neighbor thermodynamic model, which dictates that the stability of an RNA duplex depends heavily on adjacent base-pair stacks rather than isolated, independent pairs.

Because RNA strands hybridize in an anti-parallel orientation, the algorithm evaluates sequence A in the $5' \rightarrow 3'$ direction and sequence B in the $3' \rightarrow 5'$ direction. Let the matrix $C_{i,j}$ represent the minimum free energy of a duplex ending exactly with a base pair between nucleotide i of sequence A and nucleotide j of sequence B. The recurrence relation is defined as:

$$C_{i,j} = \min_{\substack{i'<i \\ j'>j}} \left(C_{i',j'} + \Delta G(i', j' \rightarrow i, j) \right) \tag{2.7}$$

where $\Delta G(i', j' \rightarrow i, j)$ represents the nearest-neighbor thermodynamic energy penalty (or reward) of forming an internal loop, bulge, or base-pair stack between the preceding anti-parallel pair (i', j') and the new pair (i, j). To prevent the algorithm from evaluating biologically implausible, excessively long gaps, the search limits the maximum allowed distance between interacting pairs, typically restricting asymmetric loop sizes to a predefined constant (e.g., 30 nucleotides).

The final output of RNAduplex is the optimal local interaction site and its associated hybridization energy. While extremely fast, the core limitation of the algorithm stems directly from its mathematical simplification: because it cannot model intramolecular folding, it may predict highly stable intermolecular interactions in regions of the RNA that are, in reality, biologically inaccessible due to strong competing internal secondary structures. The parameters used where the same as Zacharopoulou *et al.* 2024.

IntaRNA IntaRNA in its simplest mode can be used to scan genomes for target sites of specific small RNAs using a heuristics based approach that can be modified via a different number of command line arguments. The next mode in terms of simplicity is called IntaRNA exact and it aims to minimize the following sum:

$$E_{hybrid}(I) + ED1(I) + ED2(I) \tag{2.8}$$

In this model, E_{hybrid} denotes the hybridization energy of the base pairs forming between the two distinct molecules. Meanwhile, E_D represents the accessibility energy, the energy required to break existing internal (intra-molecular) bonds so the target sequences can interact. The model evaluates specific inter-molecular binding patterns, such as helical stackings, bulges, and interior loops. Because internal base pairings are not strictly mapped out, IntaRNA's predictions are flexible enough to account for any potential structural arrangement during single-site interactions. The ensemble mode of IntaRNA is based on a partition function computation that aims to find the optimal target site among many on the target sequence such that:

$$\text{arg max}\, Z(S) = \exp\left(-\frac{E_{D1}(S)}{RT}\right)\exp\left(-\frac{E_{D2}(S)}{RT}\right)\sum \exp\left(-\frac{E_{hybrid}(I)}{RT}\right) \tag{2.9}$$

The site-specific partition function, $Z(S)$, is calculated by multiplying the Boltzmann weights of the accessibility penalties (E_{D1} and E_{D2}) by the sum of the Boltzmann weights for all hybridization interactions (I) at that site. Here, R represents the universal gas constant and T is the temperature (with $RT \approx 0.616$ at 37°C). This partition function is used to determine the ensemble energy of all interactions at a given site using the equation $E(S) = -RT\,\ln(Z(S))$. By calculating ensemble energy rather than relying on a single, static base-pairing model, this approach accounts for the inherent flexibility and dynamic nature of the interacting regions. Furthermore, summing the partition functions across all potential sites yields the global partition function, $Z_{all} = \sum Z(S)$. This allows for the calculation of the probability that a specific interaction I will form, defined as $P_E(I) = \exp(-E(I)/RT)/Z_{all}$. IntaRNA provides additional options that function like other widely used software for the same task like RNADuplex, TargetScan and RNAup (Lorenz, Bernhart, *et al.* 2011; Hofacker *et al.* 1994; Lorenz, Hofacker, *et al.* 2016; Turner & Mathews 2010).

The command that was used to run IntaRNA is the following, where these options allow for rather relaxed parameters in terms of feasible duplex energy values, in order to permit feature extraction for negative samples as well (outDeltaE and outMaxE), the acc parameters force the model to calculate global accessibility for intramolecular interactions that are one of its benefits from RNADuplex. The other parameters are necessary based on documentation in order to get the appropriate model (P or X) to function as intended.

```
$ IntaRNA --mode=M --accW=0 --accL=0 --outDeltaE=100 --
    outOverlap=B --model=P(Ensemble) or X(Exact) --noseed
    --outMaxE=100 --outNoLP
```

Conservation

An important aspect that can many times determine if a sequence is functional is how well conserved it is evolutionarily. In essence, it is expected that many MREs will be more conserved across species compared to other non regulatory sequences, with the exception of species-specific ones. The two most widely used ways to measure conservation of nucleotide sequences are via Phastcons (Yang 1995; Felsenstein & Churchill 1996;

Mayor *et al.* 2000; Margulies *et al.* 2003; Siepel, Bejerano, *et al.* 2005; Siepel & Haussler 2005) and PhyloP (Pollard *et al.* 2010). Phastcons employs a two-state phylogenetic hidden Markov model (phylo-HMM) to calculate a conservation score for each nucleotide, which represents the posterior probability that the base belongs to a conserved element rather than a non-conserved one. There are two files one can download and use that have genome-wide conservation scores inside, genePhastCons100way and genePhastCons470way. The former was constructed taking into account 100 vertebrate species, while the latter 470 mammalian species. PhyloP scores measure evolutionary conservation at individual alignment sites, indicating whether a site is evolving slower (positive scores) or faster (negative scores) than expected under a neutral model of evolution.

In the scope of this thesis, Phastcons100way and PhastCons470way were used and the results were compared. Since these scores are sequential in nature, features have to be engineered by constructing metrics that measure local properties of the distribution of these scores. On that end, the MRE sequences were split in three segments (3', central and 5' and seed) and a number of statistical features were calculated for each region.

Sequence

The nucleotide sequence itself is arguably the hardest one to draw tabular features from. One could start in a rudimentary way by calculating local AU and GC percentages, however this only works in for specific regions and on specific types of interactions (e.g AU percent flanking upstream of the seed region). For that reason, the approach that was followed aimed to use the nucleotides as a proxy for creating a more detailed energy profile of the sequences. By mapping dinucleotides to their stacking energy values according to Turner & Mathews 2010, it became feasible to create a numerical vector of energy scores that are based on experimental data and that can be used to extract features describing their distributions in specific subregions of each sequence. In total, over 100 such features were extracted from both the MRE and the miRNA by splitting them to 6 regions each before calculating the features. Only a portion of them made it to the final model, using an advanced feature selection method (see Feature Selection). Trimer frequencies and related statistical features were also extracted but by grouping the nucleotides to Purines and Pyrimidines beforehand in order to reduce the dimensionality of the resulting table.

Feature Selection

After training and evaluating autogluon on a total of 190 features, the featurewiz library was utilized for feature selection (Seshadri 2020). Featurewiz employs an advanced feature selection algorithm that is based on the Minimum Redundancy Maximum Relevance (MRMR) scheme, which ensures that the features with the maximum defined relevance score to the desired variable are selected while at the same time, ones that are correlated with each other are excluded. To achieve this, it uses the SULOV method (Searching for Uncorrelated List of Variables), and a recursive XGBoost model that selects the best subset of features after SULOV. All of the steps are described below:

1. Find all variables that are correlated with Pearson Correlation Coefficient > 0.7
2. Find the Mutual Information score of each one with the target variable
3. From the correlated pairs eliminate the features with lower MI with the target
4. Split the data for kfold cross validation before passing them to the XGBoost model
5. For each fold, find the top features on train using validation for early stopping
6. Combine all sets of features from each fold and deduplicate them

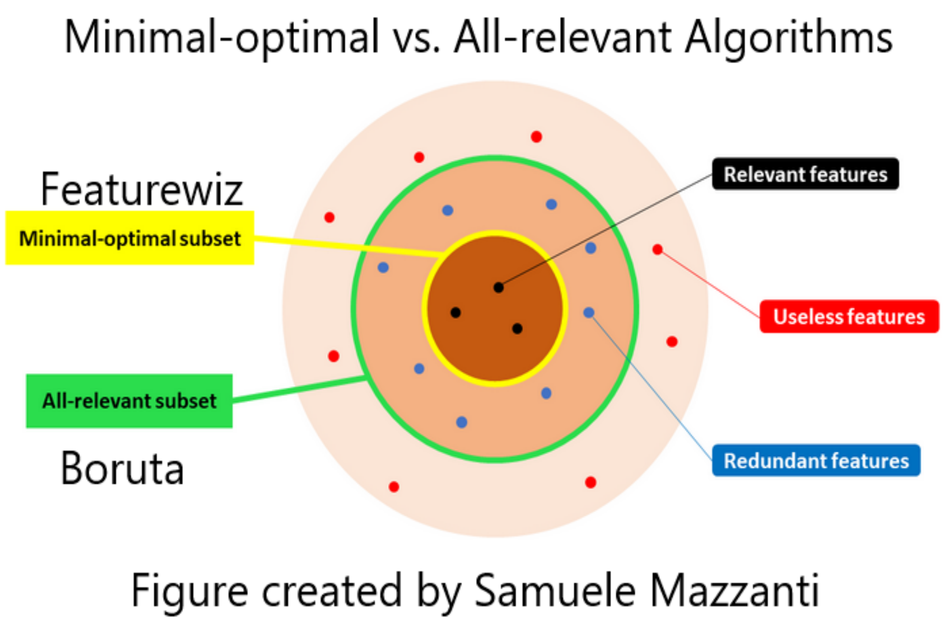


Figure 2.2: MRMR vs Boruta. (Seshadri 2020)

2.3.2 Language Models

When it comes to Language Models, the need for feature engineering is minimal as the only input to be processed is typically the nucleotide sequence after some sort of encoding (e.g One Hot Encoding or Kmer Encoding). The most intensive part of developing such models is finding the appropriate architecture that, at least in some aspect, fits the structure of the problem to be solved and performing hyperparameter tuning afterwards. The caveat that comes due to the nature and complexity of these models however, is the reduced interpretability in terms of results. Interpreting the decisions of such models is still an active area of research and although specific methods have been developed (such as Shapley values M. Li *et al.* 2024), a lot of involvement is still necessary for comprehending their meaning depending on the task at hand.

2.4 Feature Importance

Finally, the output of the pipeline contains two ways to interpret the model's predictions:

- Global Permutation based score for each feature
- Per Sample Shapley Value for each feature

2.4.1 Global Interpretability: Permutation Feature Importance in AutoGluon

While highly predictive, the stacked, multi-layer ensemble models generated by AutoGluon are inherently opaque. To extract global interpretability without relying on a specific underlying model architecture, AutoGluon natively implements Permutation Feature Importance (PFI).

Originally introduced in the context of Random Forests, PFI measures the structural reliance of a trained model on a specific feature by observing the degradation in predictive performance when that feature's information is systematically destroyed.

Mathematically, let M be the trained AutoGluon predictor, $D = \{(X_i, y_i)\}_{i=1}^{N}$ be the validation dataset, and $s(M, D)$ be the baseline evaluation score (e.g., Accuracy, F1, or negative RMSE). To calculate the importance of a specific feature j, its values are randomly shuffled across all N instances to create a corrupted dataset, $\tilde{D}_j$. This breaks any true relationship between feature j and the target y without altering the feature's marginal distribution. The permutation feature importance, I_j, is defined as the resulting drop in performance:

$$I_j = s(M, D) - s(M, \tilde{D}_j) \tag{2.10}$$

A high I_j indicates that the model relies heavily on feature j, as disrupting its distribution severely penalizes the score. Conversely, an I_j near zero suggests the feature is either irrelevant or highly correlated with other features in the ensemble that compensate for its absence.

2.4.2 Local Interpretability: SHapley Additive explanations (SHAP)

While permutation importance provides a macroscopic view of feature relevance, it fails to explain individual predictions. To achieve local interpretability, modern machine learning relies on SHapley Additive exPlanations (SHAP), a framework grounded in cooperative game theory.

SHAP frames the prediction of a single instance as a cooperative game where the ”payout” is the model’s output, and the ”players” are the input features. The goal is to distribute the difference between the actual prediction and the base expected value fairly among all features.

For a given model f and a specific data instance x, let F be the set of all features. The Shapley value ϕ_i for feature i is calculated by computing its marginal contribution across every possible permutation of feature subsets. Mathematically, it is formulated as:

$$\phi_i(f, x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!\,(|F| - |S| - 1)!}{|F|!} \left[f_x(S \cup \{i\}) - f_x(S) \right] \tag{2.11}$$

where S is a subset of features not containing i, and $f_x(S)$ represents the model’s expected prediction when only the features in subset S are known.

The term in the brackets isolates the exact marginal impact of adding feature i to the existing coalition S. Because the order in which features are added changes their marginal contribution (due to feature interactions), the combinatorial fraction acts as a weighting factor. It ensures that the contribution is averaged over all possible sequences in which the coalition could be formed.

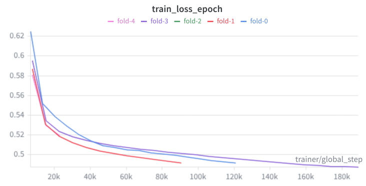
The defining theoretical advantage of SHAP is local accuracy (additivity): the sum of all feature contributions ϕ_i exactly equals the difference between the model’s output for instance x and the global expected prediction. This provides a mathematically rigorous guarantee of fair attribution that heuristic explanation methods lack.

3. RESULTS

3.1 Language Models

Inspired by microT-CNN (Zacharopoulou *et al.* 2024), two approaches were followed with the final goal of making an enhanced algorithm for achieving site-level miRNA target site prediction. Language model architectrures were developed based on Mamba in order to test their capabilities of learning miRNA binding rules based only on nucleotide sequences. The initial experiment involved using the kmer encoded (k=4) chimeric sequences, that are the result of merging the miRNA and MRE sequences as input to a Transformer model. Unfortunately, this led to a model that seemed to be underfitting despite subsequent trials to fine tune its hyperparameters. The same held true even with the attempt of making a siamese neural network that processed the miRNA and the MRE sequences separately and tried to minimize the distance between them in the embedded space. The Mamba architecture seemed to yield similar results and thus a change in terms of encoding was needed.

The proposed Mamba architecture functions as an end-to-end binary classifier for chimeric RNA sequences. Initially, the input sequences, represented as 5-dimensional one-hot encoded tokens (accounting for the four nucleotides and a segment identifier), are projected into a continuous embedding space of dimension $d_{\text{model}} = 32$ using a dedicated sequence encoder with $batch_size = 256$. The core of the network consists of a stacked sequence encoder comprising N residual blocks (by default, $N = 2$). Each block follows a pre-normalization design, sequentially applying Layer Normalization, a Mamba-2 state-space layer, and dropout regularization ($p = 0.3$) before being added to the residual connection. To aggregate the temporal representations, the output of the final Mamba block is passed through a terminal Layer Normalization and an attention-based pooling layer. This pooling mechanism dynamically weights and reduces the sequence length to a single fixed-size vector while explicitly masking out padded tokens. Finally, a linear classification head projects the pooled representation down to a single scalar logit, which is optimized via a binary cross-entropy objective. Kmer encoding was swapped with One Hot encoding which contains a fifth position with a label for whether the sequence belongs to the MRE or the miRNA and the Mamba architecture was evaluated. A grid of Mamba's performance across epochs can be seen below where different metrics are depicted for the Mamba architecture. More specifically, each line represents a model instance running on a different, miRNA family split, fold of the data utilizing kfold cross validation:



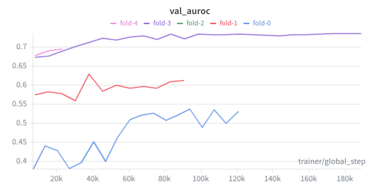
(a) Train Loss

(b) Train APS

(c) Train AUROC

(d) Validation Loss

(e) Validation APS



(f) Validation AUROC

Figure 3.1: Comprehensive grid of results showing Mamba Performance

From the results it can be surmised that Mamba struggles to generalize to new miRNA families in this assumed architecture as the performance of each fold in the validation sets differs significantly, while the average performance of the model instances in terms of APS on the test and leftout sets is around 0.67 after increasing regularization. Optuna (Akiba *et al.* 2019) was used in order to perform tuning for 5 different hyperparameters, setting the Validation set APS as the score to maximize:

1. Learning Rate (1×10^{-5} to 1×10^{-2}), Best: 2×10^{-3}
2. Weight Decay (1×10^{-6} to 1×10^{-2}), Best: 2×10^{-4}
3. Hidden Dimensions (32 to 128), Best: 32
4. Number of Mamba Layers (1 to 3), Best: 3
5. Dropout (0.1 to 0.3), Best: 0.15, changed to 0.3 to mitigate overfitting

Additionally, another configuration for the Mamba architecture was tested where Bidirectional Mamba blocks were used to read each sequence and its reverse complement. Unfortunately, this did not improve the performance and hyperparameter tuning remains to be attempted for this model.

3.2 Autogluon

Given that Mamba models were overfitting even with few parameters and regularization, a logical option would be to attempt using simpler machine learning algorithms, which is why Autogluon was employed. The initial feature set contained everything mentioned in the Methods chapter but the only energy related feature was the Molecular Free Energy value output by RNADuplex, while any duplex related feature comes from its prediction for the duplex and sequence related features are limited to local AU/GC percentages. Additionally this is the only feature set where Contrafold related features appear as it was discovered that removing them had no effect on the performance of the model.

Table 3.1: RNADuplex Performance

RNADuplex	Training	Validation	Test	Holdout
ROC-AUC	76.8	76.3	75.6	75.6
APS	79.5	78.6	79.6	79.8

The first improvement arose after the decision to use IntaRNA instead of RNADuplex. Due to the fact that IntaRNA takes into consideration intramolecular pairing and provides multiple Energy features instead of just an MFE value like RNADuplex, it is a great candidate for improving model performance. Initially, the Exact mode of it was used and indeed lead to a significant performance boost.

Table 3.2: IntaRNA Performance

IntaRNA	Training	Validation	Test	Holdout
ROC-AUC	79.1	78.8	78.4	79
APS	82.1	81.5	82.3	83.3

Given the increase in performance, an attempt was made to incorporate the ensemble mode of IntaRNA. The issue with this mode is the absence of a clear duplex structure which occurs due to its algorithm. Therefore, both modes were used and only sites of interaction predicted by both were kept. This allows for retaining the ensemble mode calculated energy features and a duplex structure for extraction of binding type information. The results of this feature set are shown below:

Table 3.3: IntaRNA Ensemble Performance

IntaRNA Ensemble	Training	Validation	Test	Holdout
ROC-AUC	79.8	79.2	78.7	79.4
APS	82.6	81.9	82.6	83.6

Maintaining the features provided by the combined output of Exact and Ensemble modes, extra features were added that were based on the nucleotide sequence, not restricted to the seed or binding site predictions of IntaRNA. The sequence was read in dinucleotides, each one was mapped to an energy value as described in Methods and statistical features were extracted. Additionally, expression values for each miRNA sequence were extracted from Manakov *et al.* 2022 and used as a feature with the hypothesis that they would help the model learn the dependency between expression levels and interactions. Unfortunately, expression levels did not increase the performance of the model.

Table 3.4: IntaRNA Ensemble with expression and sequence features Performance

IntaRNA Ensemble with expression and sequence	Training	Validation	Test	Holdout
ROC-AUC	82	80.4	80.4	80.5
APS	84.4	82.7	83.8	84.4

Since expression alone did not account for experimental and tissue specific biases, correction of the dataset to account for false negatives was performed utilizing TarBase and weights were added to each samples following the aforementioned formula. Metrics were reported taking into account these weights so as to express the model's generalization to new families more reliably, and a non-weighted APS value is also reported for direct comparison with previous feature sets.

Table 3.5: IntaRNA Ensemble Corrected Performance

IntaRNA Ensemble Corrected	Training	Validation	Test	Holdout
APS	84	83.7	83.4	84
ROC-AUC (weighted)	88.1	81.6	81.9	77.6
APS (weighted)	89.5	84.1	85.2	81.6

In order to perform a proper comparison for the goal of miRNA target prediction, PhastCons100way vectors were used instead of PhastCons470way while keeping the rest of the feature set the same. The results show negligible changes in performance and thus the features from PhastCons470way were kept since they are mammal specific.

Table 3.6: IntaRNA Ensemble PhastCons100way

IntaRNA Ensemble PhastCons100way	Training	Validation	Test	Holdout
APS	84.1	83.7	84.1	84.1
ROC-AUC (weighted)	89.3	81.6	82.2	77.9
APS (weighted)	90.5	84.1	85.6	81.8

Energy and duplex related features were extracted from scratch with the addition of a parameter in IntaRNA that disables the accessibility computation for miRNA sequences, given that they are bound by AGO proteins and are therefore not folded while additional conservation features were added describing more aspects of the distribution of these scores locally across the sequence of the MRE. Feature selection was performed via Featurewiz which led to a reduction of 190 features in total to 58 with minimal losses in performance.

Table 3.7: IntaRNA Ensemble Featurewiz

IntaRNA Ensemble Featurewiz	Training	Validation	Test	Holdout
APS	83.5	83.4	83.7	83.6
ROC-AUC (weighted)	88.2	81.3	81.9	77.1
APS (weighted)	89.4	83.8	85.2	81.3

Finally, the model was trained on the entirety of the training dataset and tested once again on the two test sets. These results show that the model is able to learn to distunguish Binding from non Binding with metrics close to SOTA models for the same task. Furthermore, evaluating it on metrics weighted on miRNA families shows something that other SOTA models have not examined thus far which is a more detailed description of the model to generalize to new miRNA families not only via testing on unseen data but by also reflecting the task within model training and metrics.

Table 3.8: Final Model

Final Model	Training	Validation	Test	Holdout
APS	85	83.1	84	83.7
ROC-AUC (weighted)	92.9	81.4	82.3	77.1
APS (weighted)	93.8	84.3	85.6	81.4

Table 3.9: Final Model vs miRBind vs TargetScan

Evaluation	TargetScan	miRBind	miRBind2	Final Model
Manakov Test	77	84	85	84
Manakov Leftout	76	81	83	83

Table 3.10: Features extracted by the unified feature-extraction pipeline. Notation: E $= (E_1, \ldots, E_n)$ denotes the vector of Turner nearest-neighbour stacking energies (kcal mol^{-1}, 37 °C) for consecutive dinucleotides of a given sequence region; $\Delta E_i = E_{i+1} - E_i$. The duplex vector v encodes per-position alignment codes (`1` = match, `2` = mismatch, `3` = MRE bulge, `4` = miRNA bulge).

Feature	Definition / Formula
IntaRNA thermodynamic energies	
`Eall`	Ensemble energy of all considered interactions (-RT*log(Zall)). Computed by IntaRNA.
`Eall1`	Ensemble energy of all considered intra-molecular structures of the Target sequence (given its accessibility constraints). Computed by IntaRNA.
Duplex-structure statistics	
`total_matches`	$\sum_i \mathbf{1}[v_i = 1]$ (number of Watson–Crick + GU matched positions).
`total_mre_bulges`	$\sum_i \mathbf{1}[v_i = 3]$ (unpaired MRE nucleotides opposite a miRNA match).
`total_gu_wobbles`	Total G : U wobble pairs across the full duplex.
Seed region (miRNA positions 2–8)	
`seed_matches_2_8`	Number of matches (`1`) in the duplex vector at miRNA positions 2–8.
`gu_wobbles_in_seed_2_8_pos`	Number of G : U wobble pairs among seed-region matched positions.
`seed_au_content`	Fraction of seed matched pairs that are A : U: $n_{\text{AU}}/n_{\text{seed pairs}}$.
Non-seed / 3′ supplementary region (miRNA pos. $\geq$ 9)	
`consecutive_matches_minus_seed`	Longest run of consecutive matches (`1`) in the non-seed portion of the duplex vector.
`effective_3prime_matches`	Non-seed matched pairs minus G : U wobble pairs in the non-seed region: $n_{\text{nonseed pairs}} - n_{\text{GU}}^{\text{(nonseed)}}$.
Conservation features (phastCons)	
`seed_conservation_median`	Median phastCons score over MRE positions aligned to miRNA seed (positions 1–7).
`seed_conservation_max`	Maximum phastCons score over the same seed-aligned MRE positions.
`seed_conservation_min`	Minimum phastCons score over the same seed-aligned MRE positions.
`five_prime_flank_conservation_mean`	Mean phastCons of the k nucleotides immediately 5′ of the seed MRE region ($k =$ `flank_size`).
`three_prime_flank_conservation_mean`	Mean phastCons of the k nucleotides immediately 3′ of the seed MRE region.

continued on next page

Table 3.10 continued

Feature	Definition / Formula
`conservation_range`	max($\mathbf{c}$) − min($\mathbf{c}$), where **c** is the full 50-nt conservation vector.
`downstream_max_consecutive_drop`	Maximum consecutive drop in phastCons scores within the downstream third of the conservation vector **c**.
`downstream_skewness`	Skewness of the phastCons scores within the downstream third of the conservation vector **c**.
`downstream_gini`	Gini coefficient of the downstream third of **c**: $G = \frac{2\sum_i i\, c_{(i)}}{n\sum c_{(i)}} - \frac{n+1}{n}$, with $c_{(i)}$ sorted.
Thermodynamic / sequence features — miRNA 3′ region (positions > 8)	
`mirna_3p_energy_mean`	$\bar{E} = \frac{1}{n}\sum_{i=1}^{n} E_i$.
`mirna_3p_energy_std`	$\sigma = \sqrt{\frac{1}{n}\sum(E_i - \bar{E})^2}$.
`mirna_3p_energy_min`	min(E_i).
`mirna_3p_energy_max`	max(E_i).
`mirna_3p_energy_asymmetry`	$\bar{E}_{5'\text{-half}} - \bar{E}_{3'\text{-half}}$, where the energy vector is split at the midpoint.
`mirna_3p_energy_gradient`	Pearson correlation r between position index and energy: $r(\mathbf{x}, \mathbf{E})$, $x_i = i$.
`mirna_3p_5p_terminal_energy`	E_1 (energy of the first dinucleotide).
`mirna_3p_3p_terminal_energy`	E_n (energy of the last dinucleotide).
`mirna_3p_energy_volatility`	Mean absolute step change: $\frac{1}{n-1}\sum\lvert\Delta E_i\rvert$.
`mirna_3p_energy_max_jump`	max$\lvert\Delta E_i\rvert$.
`mirna_3p_stability_run_frac`	Fraction of steps with $\lvert\Delta E_i\rvert < 0.3\,\text{kcal mol}^{-1}$.
`mirna_3p_energy_oscillation`	Fraction of sign changes in the non-zero elements of sgn($\Delta\mathbf{E}$).
`mirna_3p_energy_drift`	$E_n - E_1$.
`mirna_3p_YYY_freq`	Frequency of pyrimidine triplets (Y = C | U): $n_{\text{YYY}}/n_{\text{trinuc}}$.
`mirna_3p_YYR_freq`	Frequency of YYR trinucleotide pattern (R = A | G).
`mirna_3p_YRY_freq`	Frequency of YRY trinucleotide pattern.
`mirna_3p_YRR_freq`	Frequency of YRR trinucleotide pattern.
`mirna_3p_RYY_freq`	Frequency of RYY trinucleotide pattern.
`mirna_3p_ggg_count`	Count of `GGG` occurrences in the 3′ region subsequence.

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Table 3.10 continued

Feature	Definition / Formula
Thermodynamic / sequence features — miRNA seed region (positions 2–8)	
`mirna_seed_energy_mean`	$\bar{E}$ of the seed dinucleotides.
`mirna_seed_energy_min`	$\min(E_i)$ over the seed.
`mirna_seed_energy_range`	$\max(E_i) - \min(E_i)$ over the seed.
`mirna_seed_energy_asymmetry`	$\bar{E}_{5'\text{-half}} - \bar{E}_{3'\text{-half}}$ within the seed.
`mirna_seed_5p_terminal_energy`	E_1 of the seed.
`mirna_seed_energy_volatility`	Mean absolute step change: $\frac{1}{n-1}\sum\lvert\Delta E_i\rvert$ within the seed.
`mirna_seed_RRR_freq`	Frequency of purine triplets (R = A | G): $n_{\text{RRR}}/n_{\text{trinuc}}$.
`mirna_seed_RRY_freq`	Frequency of RRY trinucleotide pattern in the seed.
Thermodynamic / sequence features — MRE 3′ half	
`mre_3p_energy_mean`	$\bar{E}$ of the 3′ half of the MRE dinucleotides.
`mre_3p_energy_range`	$\max(E_i) - \min(E_i)$.
`mre_3p_energy_max_jump`	$\max\lvert\Delta E_i\rvert$.
`mre_3p_unique_trinuc_ratio`	$\lvert\{\text{unique trinucleotides}\}\rvert \,/\, \min(n_{\text{trinuc}}, 64)$.
Thermodynamic features — MRE 5′ half	
`mre_5p_energy_mean`	$\bar{E}$ of the 5′ half of the MRE dinucleotides.
`mre_5p_energy_range`	$\max(E_i) - \min(E_i)$.
`mre_5p_energy_max_jump`	$\max\lvert\Delta E_i\rvert$.
Binding-type indicator variables (one-hot)	
`binding_type_5mer.mismatch.3prime`	$\mathbf{1}[\text{type} = \text{5mer with mismatch and 3′ supplementary pairing}]$.
`binding_type_6mer.mirna.bulge.3prime`	$\mathbf{1}[\text{type} = \text{6mer with miRNA-side bulge and 3′ pairing}]$.
`binding_type_3prime.compensatory`	$\mathbf{1}[\text{type} = \text{3′ compensatory (}\geq\text{8 effective 3′ matches)}]$.
`binding_type_6mer.mismatch.3prime`	$\mathbf{1}[\text{type} = \text{6mer with mismatch and 3′ supplementary pairing}]$.
`binding_type_seedless`	$\mathbf{1}[\text{type} = \text{seedless (no canonical seed match)}]$.

In order to interpret the predictions during inference, the model was used to predict samples in the leftout dataset and the calculation of Shapley values followed. In order to avoid calculating for thousands and possibly millions of samples depending on the dataset, K-Means is used in order to group the samples within the feature space in one of 25 clusters. Then, instead of calculating the Shapley value using all of the different sample combinations as background (thus performing background calculation as many times as samples), it does so, based on the number of cluster representatives. A list of the top 30 features ranked based on the mean of their importance across 200 samples is shown below as well as PFI based on all of the samples of the same file:

Table 3.11: Global Feature Importance (Mean Absolute SHAP Values) - Complete Results

Rank	Feature	Mean Abs SHAP
1	Eall	0.150368
2	seed_matches_2_8	0.112795
3	gu_wobbles_in_seed_2_8_pos	0.033197
4	mre_5p_energy_mean	0.019582
5	total_mre_bulges	0.018676
6	downstream_gini	0.014208
7	Eall1	0.013175
8	binding_type_6mer.mismatch.3prime	0.011076
9	seed_conservation_min	0.010525
10	total_gu_wobbles	0.010238
11	mre_3p_energy_mean	0.009645
12	consecutive_matches_minus_seed	0.008837
13	conservation_range	0.007518
14	five_prime_flank_conservation_mean	0.005568
15	seed_conservation_median	0.005387
16	mirna_3p_energy_mean	0.005385
17	binding_type_5mer.mismatch.3prime	0.004369
18	mirna_seed_energy_mean	0.003605
19	binding_type_3prime.compensatory	0.003533
20	total_matches	0.002513
21	mirna_seed_energy_min	0.002442
22	mre_5p_energy_max_jump	0.002433
23	mirna_seed_RRY_freq	0.002424
24	seed_au_content	0.002269
25	seed_conservation_max	0.002258
26	mre_3p_energy_max_jump	0.001799
27	binding_type_6mer.mirna.bulge.3prime	0.001600
28	mirna_3p_YYR_freq	0.001405
29	downstream_skewness	0.001267
30	mirna_3p_YRY_freq	0.001201

The features that seem to dominate the predictions are the Duplex features with the most important one being the Ensemble energy of the duplex. That is to be expected since the use of IntaRNA alone was the driver for the most prominent increase in metrics. That is also evident given that within the top 10 features, many are based on the binding type and the matches predicted within the seed, while energy features that are based on Turner stacking energies are also important.

Table 3.12: Permutation Feature Importance - Complete Results

Rank	Feature	Permutation Importance	Std Dev
1	Eall	0.081164	0.006075
2	seed_matches_2_8	0.051036	0.005536
3	gu_wobbles_in_seed_2_8_pos	0.012984	0.003629
4	total_mre_bulges	0.004732	0.002487
5	Eall1	0.004083	0.002012
6	mre_5p_energy_mean	0.003394	0.001923
7	binding_type_3prime.compensatory	0.003389	0.003181
8	mre_3p_energy_mean	0.003222	0.001212
9	binding_type_seedless	0.002955	0.001255
10	consecutive_matches_minus_seed	0.002834	0.001218
11	total_gu_wobbles	0.002517	0.001481
12	mirna_seed_energy_min	0.002209	0.000908
13	binding_type_6mer.mismatch.3prime	0.002101	0.000903
14	seed_au_content	0.001746	0.000644
15	mirna_seed_energy_volatility	0.001266	0.000481
16	mirna_3p_energy_min	0.001246	0.000517
17	downstream_skewness	0.001226	0.000381
18	mirna_seed_energy_mean	0.001180	0.000564
19	mirna_seed_RRY_freq	0.001130	0.000510
20	total_matches	0.000951	0.000741
21	binding_type_5mer.mismatch.3prime	0.000936	0.000776
22	mirna_seed_5p_terminal_energy	0.000882	0.000451
23	mirna_3p_YYY_freq	0.000870	0.000418
24	effective_3prime_matches	0.000780	0.000549
25	mirna_3p_energy_std	0.000571	0.000331
26	mre_3p_energy_max_jump	0.000444	0.000350
27	mirna_3p_energy_volatility	0.000368	0.000294
28	mirna_seed_energy_range	0.000187	0.000211
29	binding_type_6mer.mirna.bulge.3prime	0.000165	0.000232
30	mre_3p_energy_range	0.000164	0.000156

Permutation Feature Importance seems to paint the same picture as IntaRNA based features dominate the top of the list as well validating that it was a sound choice to include such features for the target site prediction task. Unfortunately, although there was an attempt to extract sequence based features, due to the constraint that Machine Learning models outside of Deep Neural Networks need tabular data, some information is bound to have been lost. Since more advanced architectures can easily read such sequential data, it would be interesting as a future endeavor to incorporate sequence along with energy related features and train a more advanced model with the goal of evaluating if such features are already learned by sequence alone from such architectures.

4. DISCUSSION

MicroRNAs partake in many physiological processes and their deregulation can be found within many disease states. Given their central role in post-transcriptional regulation and the cost of experimentally validating targets of these molecules, it is vital that algorithms for this task are developed so that the binding rules are elucidated across tissues and states. Recently many machine learning and Deep learning architectures aim to perform target site prediction, either on the MRE level or the gene level with varying degrees of success. Within the scope of this thesis, an MRE target site predictor was developed by examining the performance of Language models as well as Machine Learning ensembles, using the Average Precision score as the evaluation metric across unseen miRNA-MRE pairs containing both seen and unseen miRNA families for a more robust way to measure generalization. Inspired by MicroT-CNN, feature engineering for the Machine Learning ensembles utilized tools for thermodynamic feature extraction as well as structural features of the miRNA-MRE duplex. Sequence features were based on encoding nucleotides as Purines or Pyrimidines before constructing trimer frequencies in order to reduce noise and encoding dimers of nucleotides relative to their Turner stacking energies, while conservation features describing the distribution of the PhastCons470way scores were extracted from the vectors.

The approach utilizing Machine Learning ensembles seems to yield superior results, given its rich feature space. Feature importance analysis showed that thermodynamic features along with duplex related ones are the drivers for model performance and generalization to unseen data, hinting that they may hold some promise in being used also as part of larger models to help the increase prediction quality. Given the complexity of current deep learning approaches, it is difficult to interpet their predictions and understand which known properties drive the classification. In fact due to the statistical nature of these models, the most widely used way to interpret them involves SHAP values, which while shedding some light into the black box, are not always interpetable. Tree based models although having the caveat of requiring extensive feature engineering, have the advantadge of being naturally interpetable via the thresholds they set. The ensemble developed in this thesis while interpeted in the scope of SHAP values, provides the capability of using each of the models of the ensemble separately with minimal loss in performance which allows one to use, for instance, the trained LightGBM and interpeting its thresholds for interaction classification.

Future work is needed in order to test more configurations for the Mamba architecture in this task, for instance, combining pure One Hot Encoded sequences with some form of energy gating in such models is a promising direction. Expanding the dataset to include more tissues instead of relying on samples weights would improve the model, while reducing additional problems that might arise due to the sampling bias. Finally, improving the calculation of weighted metrics to be more generalizable across different sample sizes and applying them for evaluation in other models is also necessary to gauge their performance in rare miRNA families.

The Github repository containing the code and the instructions to build the environment necessary to reproduce the Results of this thesis exist at Msc-Thesis Github

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